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Departamento de Informática

Dados e Aprendizagem Automática
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Practical Exercise no. 9

Theme Clustering: K-means, K-medoids and DBSCAN

Exercise In contrast to supervised learning, in which labels are provided alongside the data, **unsupervised learning** essentially involves obtaining inferences from data sets without human intervention. The three techniques applied in this context are **K-means**, **K-medoids** and **DBSCAN**. The first two algorithms aim to group unlabelled case studies together based on their characteristics. However, while the K-means algorithm tries to minimise the distances within a cluster, the K-medoids algorithm tries to minimise the sum of the distances between each point and its respective cluster's 'medoid'. The DBSCAN algorithm, on the other hand, finds core samples of high density and expands clusters from them.

The dataset contains 777 observations and 18 columns:

- **Private** - a factor with levels "No" and "Yes", indicating whether the university is private or public
- **Apps** - number of applications received
- **Accept** - number of applications accepted
- **Enroll** - number of new students enrolled
- **Top10perc** - % of new students from the top 10% of high school classes
- **Top25perc** - % of new students from the top 25% of high school classes
- **F.Undergrad** - number of full-time undergraduates
- **P.Undergrad** - number of part-time undergraduates
- **Outstate** - out-of-state tuition
- **Room.Board** - room and board costs
- **Books** - estimated book costs
- **Personal** - estimated personal spending
- **PhD** - % of faculty with Ph.D.'s
- **Terminal** - % of faculty with terminal degrees
- **S.F.Ratio** - student/faculty ratio
- **perc.alumni** - % of alumni who donate
- **Expend** - instructional expenditure per student
- **Grad.Rate** - graduation rate

Tasks The aim of this practical exercise is to categorise a set of universities as either private or public. Based on the provided characteristics, a set of unsupervised models were selected to solve this binary classification problem: K-means, K-medoids and DBSCAN. The following tasks are included:

T1. Load the dataset and apply methods for data exploration and visualisation.

T2. Train three unsupervised learning models using K-means clustering (*sklearn.cluster.KMeans*), K-medoids (*sklearn_extra.cluster.KMedoids*) and DSCAN (*sklearn.cluster.DBSCAN*), to classify each case study as 'private institute' or 'public institute'. Note: the 'Private' attribute indicates whether each university is a private institution. For training purposes, this attribute should be removed from the dataset.

T3. Using the value of the *'Private'* attribute, evaluate the clustering performance of each model by creating a confusion matrix (`sklearn.metrics.confusion_matrix(...)`) and a classification report (`sklearn.metrics.classification_report(...)`).

T4. Based on the results obtained in **T3**, what conclusions can be drawn? In which situations do the models perform well/poorly? How can the proposed learning models be improved?

T5. Implement clustering algorithms using the dataset from the practical work and analyse the results.